UNIT 7 PROJECT

Survey, Graph, & *Analyze*

Collect real data. Make a graph. Find the story.

Name: ____________ **Date:** ________ **Period:** ________

🎯 **Goal:** Collect real data, represent it with a graph, and analyze patterns using Unit 7 concepts. You'll design a survey, ask at least 10 people, organize your results, and find what your data is telling you.

1 PART 1

Create Your Survey

You'll design a survey question and collect responses from **at least 10 people**.

Requirements:

✓ Your question must be clear

✓ Your responses must be *categories* (not open-ended)

💡 **Good examples:**

- Favorite music genre? (Pop, Rap, Country, Other)
- After-school activity? (Sports, Job, Homework, Relaxing)
- How do you get to school? (Bus, Car, Walk, Other)

Your survey question:

Your answer choices:

PART 2
Collect & Organize Data

Record your responses in this **frequency table**. Use tally marks as you collect, then count the totals.

Category	*Tally*	*Frequency*
Total		

Calculate Relative Frequency *(optional but encouraged)*:

$$\text{Relative Frequency} = \frac{\text{Category Count}}{\text{Total Responses}}$$

Category	*Frequency*	*Relative Frequency*

3

PART 3

Create a Graph

Choose **at least one** graph type to display your data:

☐ 📊 Bar Graph ☐ 🥧 Circle Graph (Pie Chart)

Must include:

- Title
- Labeled categories
- Scale (if needed)
- Accurate values

Title: ______________________

draw your bar graph here

use this for a circle graph (pie chart)

PART 4

4 Analyze Your Data

Answer in **complete sentences**.

1. Which category had the **highest** frequency?

2. Which category had the **lowest** frequency?

3. Describe the overall pattern or trend in your data.

4. Is your data **categorical** or **numerical**? Explain how you know.

PART 5

5 Make a Comparison

Ask a **second question** OR compare your results with a classmate.

Comparison question:

Answer:

5. Which dataset had a higher top category?

6. One **similarity** between the datasets:

7. One **difference**:

6

PART 6

Connect to Unit 7 Concepts

Answer briefly in your own words.

8. Does your data show **correlation**? Why or why not?

9. Would a **scatterplot** be appropriate for your data? Explain.

10. Could you draw a **line of best fit**? Why or why not?

OPTIONAL ⭐ **Stretch Challenge: Bivariate Data**

Turn your project into **bivariate data** by collecting two pieces of information per person.

Example: hours of sleep vs. test score · minutes of homework vs. screen time · height vs. shoe size.

Your bivariate question:

Create a scatterplot with both axes labeled, then describe:

scatterplot

📈 **Direction:**

positive · negative · no correlation

💪 **Strength:** ______________________________

strong · weak · none

📋 Project Rubric

Standards-based scoring · Unit 7: Statistics & Probability

Criterion	4 — Mastery	3 — Proficient	2 — Developing	1 — Beginning
Survey Design NC.8.SP – Categorical data	Question is clear, age-appropriate, with 4+ well-defined non-overlapping categories. 10+ responses.	Question is clear with 3–4 categories. 10 responses collected.	Question is somewhat unclear OR categories overlap. 7–9 responses.	Question is unclear or open-ended. Fewer than 7 responses.
Data Organization NC.8.SP.4	Frequency table is complete, accurate, and includes correct relative frequencies (decimals or percents).	Frequency table is complete and accurate; relative frequencies attempted with minor errors.	Frequency table is mostly complete with some counting errors.	Table is incomplete or has major counting errors.
Graphical Display NC.8.SP.1	Bar graph or pie chart is precise with title, labels, accurate scale, and proportional values. Visually polished.	Graph includes title, labels, and accurate values. Minor scaling issues.	Graph is present but missing a title, labels, or has visible inaccuracies.	Graph is missing or unreadable.
Data Analysis NC.8.SP.2 / .4	All four analysis questions answered in complete sentences with strong, evidence-based reasoning.	All four answered in complete sentences with reasonable reasoning.	Some answers incomplete or written as fragments.	Most answers missing or written without reasoning.
Comparison NC.8.SP.4	Identifies clear similarities AND differences between datasets with specific evidence.	Identifies one similarity and one difference with adequate explanation.	Identifies similarity OR difference but not both, OR explanation is vague.	Comparison is missing or off-topic.
Connect to Unit 7 NC.8.SP.1 / .2 / .3	Correctly identifies whether data shows correlation, scatterplot appropriateness, and line-of-best-fit suitability with justification.	Correctly answers all three questions with brief justification.	Answers 2 of 3 questions correctly OR justifications are weak.	Answers 0–1 questions correctly or skips this section.
Communication & Presentation Mathematical Practice 6	Work is neat, organized, and uses precise math vocabulary throughout.	Work is mostly neat with some math vocabulary used.	Work is readable but disorganized or lacks math vocabulary.	Work is messy, hard to follow, or lacks vocabulary.
Points Possible	4	3	2	1

Total Score: ______ / 28

Mastery: 25–28
Proficient: 19–24
Developing: 12–18 · Beginning: < 12

⭐ Optional Extension Bonus (up to +4 points)

Extension Element	*+1 each*
Bivariate question	Collected 2 pieces of data per response with a clear research question.
Scatterplot	Plotted points correctly on properly labeled axes.
Direction described	Identified positive, negative, or no correlation correctly.
Strength described	Identified strong, weak, or none with justification.